SALT'S ANTI ALIASING / FABRIC 26.3

Temporal upscaler rendering audit

FSR2, FSR3 and DLSS Super Resolution
Real game captures, vendor research and proposed fixes

24 October 2026 - America/Phoenix
Windows / Vulkan / NVIDIA RTX 3060 Ti



Test scene: isolated block silhouettes, slabs, stairs, glass and fences. The image is an actual Minecraft capture with 2x SSAA used as a spatial reference.

Main result. A shared Y-jitter convention mismatch is strongly supported by controlled game experiments. Changing only the Y jitter supplied to the native upscalers reduces instability and removes a reproduced DLSS floating-line pattern. The production implementation remains unchanged.

Separate DLSS defect. All quality modes use a 16-phase jitter cycle. That falls below NVIDIA's recommended coverage, especially at Performance and Ultra Performance.

This is a diagnostic report, not a claim that every temporal artifact can be eliminated. Proposed corrections still need validation during motion and in the packaged mod.

EXECUTIVE REVIEW

Findings and decisions

Finding	Evidence / confidence	Recommended action
S1 - Native SDK jitter Y convention	High: matched sign-only experiments across three SDK paths; DLSS detached line disappears.	Correct the SR dispatch boundary and verify moving-scene alignment.
D1 - DLSS sample coverage	Confirmed code defect: 16 phases at every scale.	Use a DLSS-specific phase count derived from input/output pixel area.
F2 - FSR2 Native AA instability	Confirmed symptom; particularly noisy thin geometry before the sign correction.	Fix the shared convention first; reassess Native AA before changing its availability.
F3 - Transparency instability	Confirmed variation on glass/fine detail. Mask generation is a plausible contributor, not a proven sole cause.	Audit masks after the primary jitter fix; replace coarse history sampling if it remains causal.
Q1 - Thin-detail loss and edge fringes	Confirmed at low input resolution. Some loss is an expected sampling tradeoff.	Keep quality presets explicit; test fidelity alongside temporal stability.
R1 - Resize and history lifecycle	Resizing recovered; all three methods signaled a reset during timed teleports.	Retain regression coverage; no crash or perpetual reset demonstrated.

Priority: correct and validate the jitter convention first. Increasing resolution, disabling masks, sharpening the image, or upgrading a DLL does not repair a coordinate mismatch.

“Confirmed symptom” means it is visible or measurable in captured game output. “Confirmed code defect” means the implementation conflicts with a documented integration requirement. A likely cause is labeled separately from both.

METHOD AND ENVIRONMENT

What was actually tested

Component	Recorded configuration
Client	Minecraft 26.3; Fabric Loader 0.19.5; Fabric API 0.162.0+26.3; Microsoft OpenJDK 25.0.2.
GPU and API	GeForce RTX 3060 Ti, 8,192 MiB; NVIDIA driver 591.86; Minecraft Vulkan backend.
Native libraries	AMD runtime contains FSR2 2.3.3 and FSR3 3.1.4 identifiers. Streamline 2.14.1; imported nvngx_dlss.dll file version 310.9.1.0.
Build state	Current development source, with pre-existing uncommitted implementation/UI work. HEAD a3025cbba534. The source hash manifest identifies the audited state; HEAD alone does not.
Isolation	A copied test world and a separate run directory. User settings and normal saves were not edited. Production source and native bridges were not changed for this audit.

Output matrix: 1280 x 720, 1920 x 1080 and 2560 x 1440. FSR2/3: Native AA, Quality, Balanced, Performance and Ultra Performance. DLSS: Quality, Balanced, Performance and Ultra Performance. The mod does not expose DLAA, so native-input DLSS was not tested.

Scene coverage: stationary opaque geometry; high-contrast silhouettes; fences and leaves; glass, water and lava; slow/fast horizontal and vertical camera sweeps; natural terrain; moving entities and chest activity; first-person hand/HUD and selected-block views; teleports and window resizing.

Capture method: full-resolution PNG readbacks from the real game render target, typically 48-96 consecutive frames after 300-500 warm-up frames. Matched phase tests use 144 frames. GPU motion-vector probes and per-dispatch input records supplement the images. These are image-quality tests, not performance benchmarks.

METHOD AND LIMITATIONS

Reading the measurements correctly

Temporal RGB SD is the standard deviation across frames at each pixel/channel, averaged over a stated region. Units use the 0-255 display-channel scale; smaller values indicate a steadier image in a frozen scene. They do not prove better detail, less blur, correct exposure, or correct geometry.

Region at 1920 x 1080	Bounds (left, top, right, bottom)
Floor	(450, 750, 1450, 1050)
Thin geometry	(1180, 400, 1420, 610)
Glass	(580, 350, 825, 595)
Leaves	(1030, 400, 1175, 530)

At other resolutions the same normalized regions are sampled. Frozen native OFF is a control: floor variation was 0.000 and thin-region variation about 0.001. Water, lava, clouds and animated content can still vary independently; the primary floor statistic avoids those surfaces.

The 2x SSAA captures use a 3840 x 2160 scene resolved to 1920 x 1080. They are a useful high-sample spatial reference, not a vendor ground-truth render. Filter differences can shift an edge centroid without moving the underlying geometry.

Camera sweeps and entity stress are driven through the game at client/server tick cadence. They stress reprojection and disocclusion, but they do not provide a synchronized, subframe-accurate reference animation. No PSNR ranking is claimed for those moving sequences.

Readback and PNG encoding perturb timing. One early diagnostic capture incurred a history reset and is excluded from the matched sign conclusions. The clean sign pairs have zero resets in the measured windows. No FPS, latency or frame-generation performance claims are based on capture-time timing.

VENDOR INTEGRATION COMPARISON

Expected output versus this mod

Contract	Expected behavior	Evidence in the mod
Sampling alignment	Submitted pixel jitter must describe the actual raster sample shift.	Current X tracks input shift; Y uses the opposite visible-image direction. Sign-only Y experiments improve all three paths.
Stationary scene	Accumulation should stabilize edges and detail; small residual shimmer can remain.	Solid edges have measurable subpixel variation; FSR2 Native AA is notably less stable before correction.
Motion	Current-to-previous vectors must match the color/depth frame and units.	Pixel-space vectors are generated; static probes are near zero and finite. Moving-scene correctness still needs stronger ground-truth verification.
Silhouettes	Sampling can soften or ring; a distinct bright line separated by a dark gap is a defect to investigate.	The isolated-block DLSS Quality capture reproduces this pattern at sharpening 0.
Reset behavior	Discontinuous camera or resolution changes must invalidate history.	A reset is observed in each method during a timed teleport, and on resize.

AMD describes the color/depth/motion/jitter contract and per-frame reset requirements for FSR2 and FSR3. NVIDIA identifies shaking, unresolved detail and fuzzy thin objects as reasons to inspect jitter and motion inputs. [A1] [A2] [N1]

Expected limitations include missing subpixel information at very low input resolutions, temporary disocclusion softness, and difficult thin/transparent surfaces. A stable image can also be excessively blurred. The report treats those tradeoffs separately from the demonstrated alignment defect.

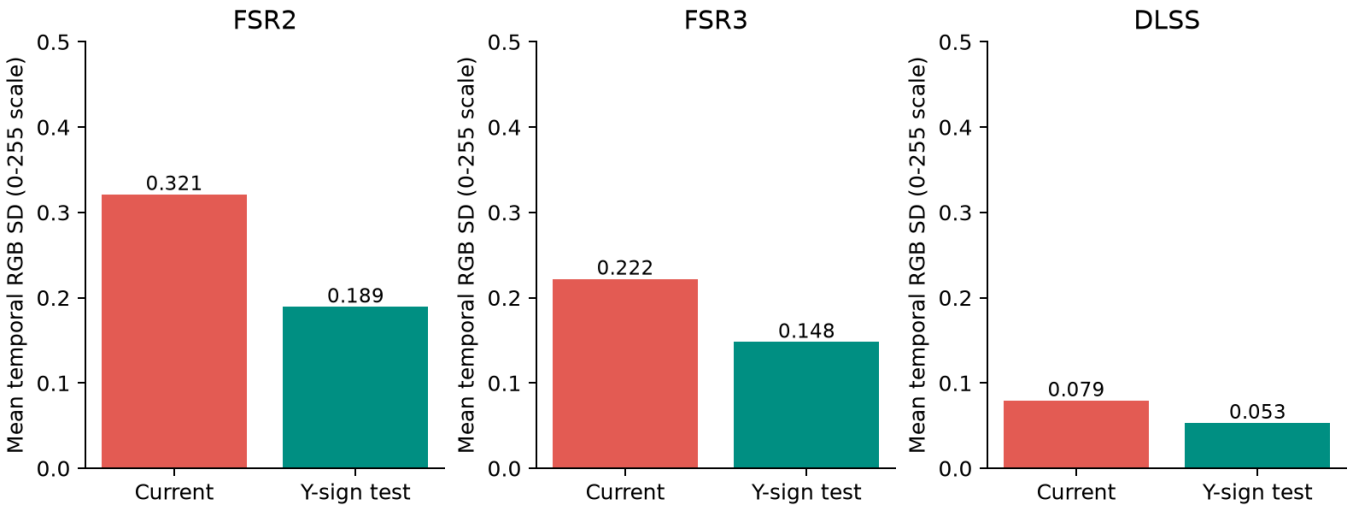
SHARED RENDERING ISSUE / HIGH CONFIDENCE

S1: shared jitter Y mismatch

The temporal controller applies clip-space offsets (2X / inputWidth, -2Y / inputHeight). Both SR bridges then submit the raw (X, Y) pair. The mod's own history-mask code separately recognizes the Y inversion: its Y jitter delta is current-minus-previous, while X is previous-minus-current.

A 48-frame input-image check measured X edge shift slopes near +1.03 pixels per submitted X jitter unit. Y edge-centroid slopes were about -0.81 to -0.85. Raster quantization makes the Y magnitude approximate; the opposing sign is the useful signal. Screenshot orientation alone is not sufficient proof, so the native-dispatch A/B test is decisive.

Quality / 1080p	Current floor SD	Y-sign test	Reduction
FSR2	0.321	0.189	41.1%
FSR3	0.222	0.148	33.4%
DLSS	0.079	0.053	33.0%



Matched 96-frame windows after 500 warm-up frames; identical floor region, sharpening 0, automatic mip bias off, no history resets. No output filtering was added.

In the isolated 1080p Quality fixture, the vertical edge-gradient centroid spans 0.371 pixels (FSR2), 0.176 pixels (FSR3), 0.297 pixels (DLSS). OFF and SSAA controls are stationary. This measures reconstructed edge movement, not physical block motion.

The experiment changes only the jitter values supplied to the SDK. It leaves the camera sample pattern, motion vectors, resolution, masks and output sharpener intact. Flipping both axes or X alone did not provide a consistent advantage. This supports a Y-convention error at the SR boundary rather than arbitrary smoothing.

SHARED RENDERING ISSUE / PROPOSED IMPLEMENTATION

How to correct the shared boundary

Proposed change: submit the raster-consistent Y sign at each native SR dispatch. Use an explicit helper or named coordinate convention, rather than an undocumented minus sign spread across call sites.

Path	Concrete location and intent
FSR2 / FSR3 SR	native/fsr_bridge/src/salts_fsr_bridge.cpp: the upscale dispatch's desc.jitterOffset. The diagnostic equivalent is (rawX, -rawY).
DLSS SR	native/dlss_bridge/src/salts_dlss_bridge.cpp: constants.jitterOffset. The diagnostic equivalent is (rawX, -rawY).
Shared temporal state	VulkanSceneTemporalController: retain a clearly documented raster-space sample pair and distinguish it from SDK-oriented values.
Masks	VulkanFsrTemporalMaskRenderer and VulkanTemporalColorRejectionRenderer: preserve their existing Y-aware reprojection convention unless a dedicated test proves it wrong.

FSR3 frame-generation caution: its presentation-input path has its own orientation conversion and already negates jitter Y for prepare. A global change to the Java FSR parameter record could double-flip that path. Apply the SR correction at the SR boundary, or make the two conventions explicit and test both. Frame generation was disabled in the new SR quality tests.

Keep motion vectors unjittered and keep scale units explicit. The pinned Streamline guide uses 1/renderWidth and 1/renderHeight for pixel vectors; the current DLSS scale follows that rule. Do not invert motion Y merely because the jitter experiment succeeded. [N2]

Before shipping: re-run vertical/diagonal camera motion, thin geometry, moving entities, camera cuts and the packaged JAR. Verify edge fidelity as well as reduced variance. This report establishes the isolated visual improvement, not completion of that shipping validation.

FSR2: instability and Native AA

FSR2 evaluates successfully in game, but stationary output is not fully stable. Native AA is the most revealing case: instability remains without resolution reduction. This supports the user's observation that changing output resolution does not remove the problem.

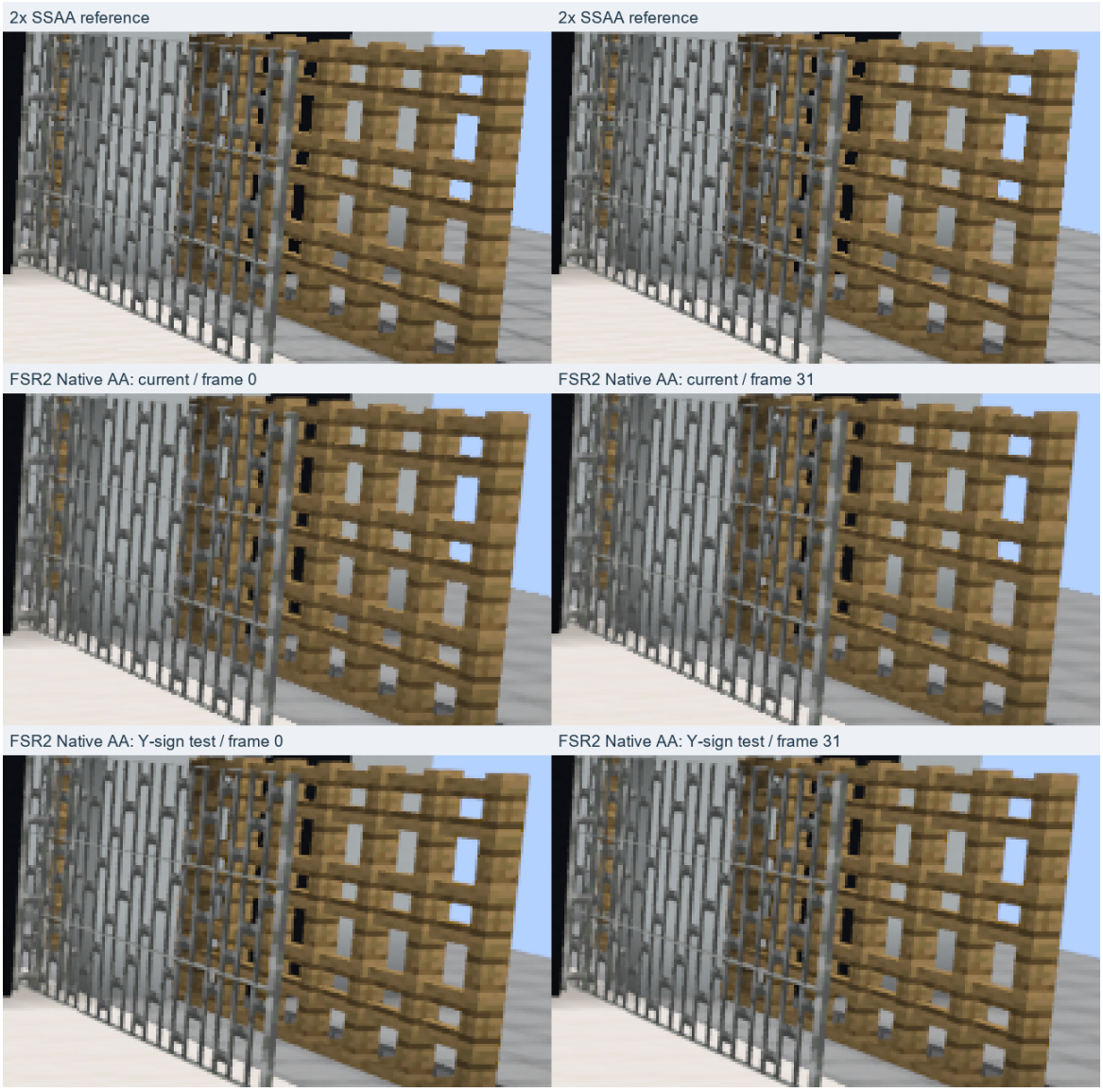
1080p preset	Floor SD	Thin SD	Glass SD
Native Aa	0.833	1.871	1.915
Quality	0.304	0.635	0.805
Balanced	0.285	0.617	0.802
Performance	0.270	0.624	0.919
Ultra Performance	0.466	1.152	1.308

These are initial unchanged-build captures. Native AA's high variation is not evidence that a lower-resolution preset is intrinsically better. Phase sequence, edge filtering, reconstruction and masking all affect this statistic.

Native-input confirmation: the matched Native AA sign test changes floor SD from 0.835 to 0.161 (80.7% lower). The correction addresses a real contributor even at full input resolution. Re-evaluate the remaining Native AA behavior after fixing the convention; do not hide the issue by silently routing FSR2 to FSR3.

FSR2's documented inputs include render-resolution depth/motion and optional reactive/composition information. Its reactive value controls reliance on the current sample, while composition information changes history protection. Those mechanisms can trade stability for responsiveness; they cannot compensate for misdescribed raster samples. [A1]

FSR2: enlarged thin-geometry evidence

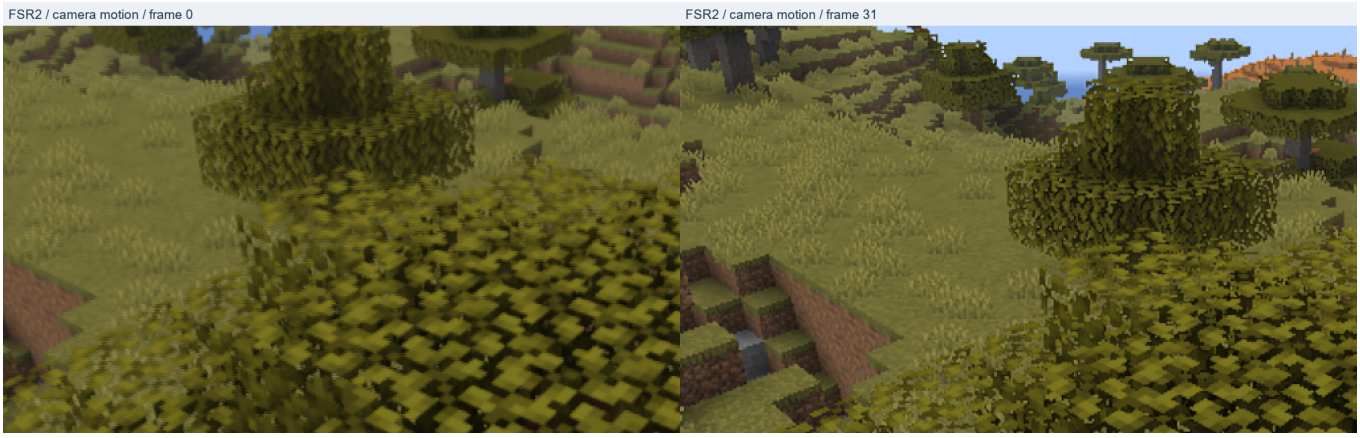


Actual game crops. Each source region is 215 x 130 pixels at 1920 x 1080; nearest-neighbor enlargement preserves pixel structure. Top: 2x SSAA reference. Middle: current FSR2 Native AA. Bottom: same Native AA with only the submitted Y jitter sign changed. Frames 0 and 31 are shown.

The thin bars and fence edges expose unstable coverage and small protruding edge pixels more clearly than a full-frame view. The experiment reduces the instability; it does not recreate detail that the scene never sampled.

Recommended FSR2 order: correct jitter orientation, retain SDK phase counts, verify depth/velocity alignment during motion, then tune masks on alpha-blended surfaces. Replacing temporal reconstruction with a blur or setting jitter permanently to zero would conceal the symptom while losing useful samples.

FSR2: motion, transparency and remedies



FSR2 Quality camera-sweep crops from natural terrain. The two frames have different camera angles and are illustrative stress views, not pixel-aligned quality references.

Fences, leaves, glass edges and newly revealed terrain are the difficult regions in the enlarged sequences. Opaque floor motion is cleaner than those features. The initial glass ROI SD is 0.805 at Quality, so output still varies after a long settle even in the frozen scene.

Disabling reactivity and composition did not remove opaque-floor shimmer. This rules out “just turn off the masks” as a complete fix. It does not prove the current masks are correct for moving transparency.

Mask implementation risk: previous samples are selected with an integer floor in fsr_temporal_masks.fsh; composition is a thresholded candidate. Fractional reprojection can toggle which history texel contributes. That is a code-based hypothesis for remaining transparency instability, not a proven cause of the detached DLSS line.

Possible solution: test depth-aware filtered history sampling, per-material alpha/reactivity, and separate debug captures of opaque/final color, depth, motion, reactive and composition buffers. Use the vendor sample as a functional comparison, then measure both trailing and shimmer on water/particles. Preserve some responsiveness; aggressively retaining all history produces ghosting.

Acceptance: static Native AA and Quality should approach the corrected diagnostic baseline; moving glass must not trade flicker for long trails. Require no new halos, missing fence segments or regressions after a teleport.

FSR3: what improves and what does not

FSR3 SR is distinct from FSR3 frame generation. All results in this section use native SR evaluation with frame generation disabled. The runtime contains the 3.1.4 algorithm identifier; these are not FSR1 spatial-upscale substitutes.

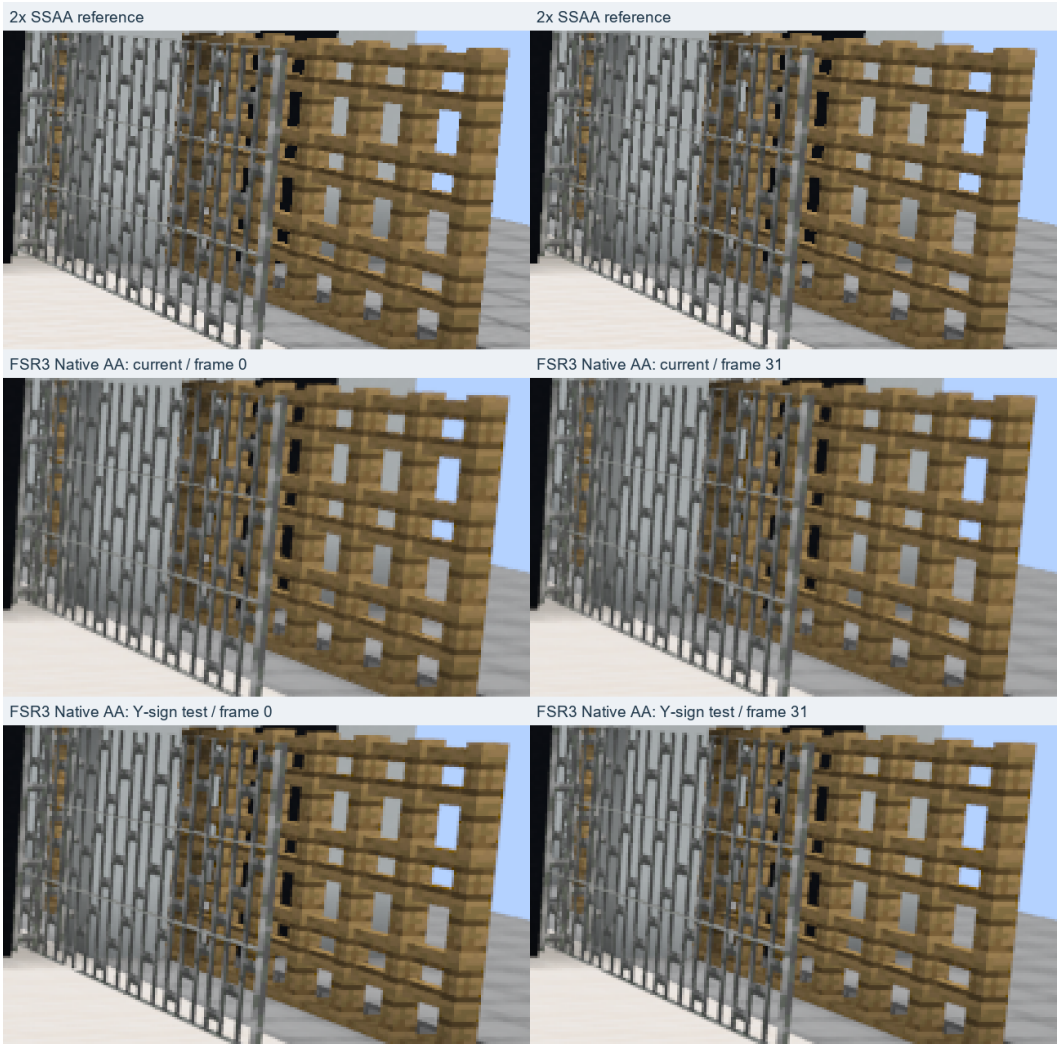
1080p preset	Floor SD	Thin SD	Glass SD
Native Aa	0.088	0.292	0.700
Quality	0.203	0.435	1.141
Balanced	0.221	0.445	1.210
Performance	0.306	0.519	1.321
Ultra Performance	0.562	1.043	1.722

Native AA is much steadier than the initial FSR2 Native AA capture. Quality also has lower floor/leaf variation than FSR2, but the glass ROI varies more. This is not a blanket ranking: different features respond differently, and low variance alone can reward blur.

Native-input confirmation: the matched FSR3 Native AA sign experiment changes floor SD from 0.090 to 0.066 (26.5% lower). The same shared convention remains relevant even where the original symptom is milder.

AMD's current upscaler guidance uses Halton jitter, scale-dependent phase coverage and explicit depth configuration. It also distinguishes the reconstruction upscaler from the interpolation pipeline. The existing SDK-query phase counts should be retained. [A2] [A3]

FSR3: native-input and glass detail



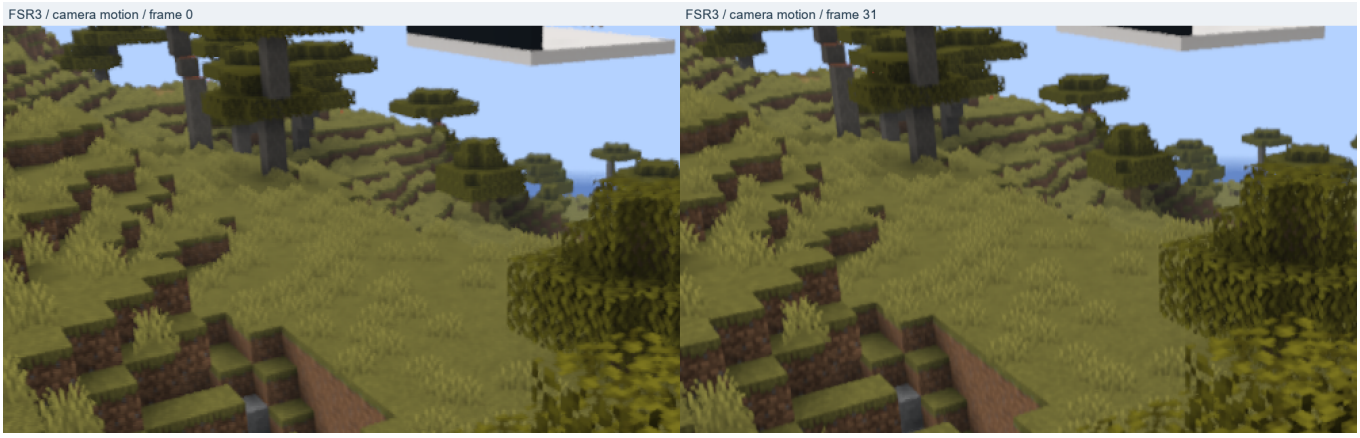
Same frozen thin-geometry region and enlargement as the FSR2 comparison. The current FSR3 Native AA path is steadier, and the Y-sign experiment provides a further check without adding a spatial blur.

At Quality, glass remains one of the higher-variation regions. The output has to reconstruct translucent edges layered over opaque surfaces that may carry a different depth/velocity. A single generic rejection heuristic is not guaranteed to describe those layers.

Solution direction: correct the shared jitter convention before adjusting mask gains. Then evaluate whether composition thresholds oscillate at static glass edges, and whether water/particles receive useful material-derived reactivity. Reducing the threshold indiscriminately can increase flicker by repeatedly discarding history.

METHOD SECTION / FSR3 FOLLOW-UP

FSR3: integration review and remaining risks



FSR3 Quality camera-sweep crops. High-frequency foliage and disocclusions remain stress cases; camera positions differ between these two frames.

Reviewed area	Finding / proposed action
Color transfer	The FSR path decodes display color to linear FP16 inputs and encodes its result back. No new systematic brightness defect was established in these tests. Keep this transfer separate from DLSS's LDR path.
Depth	Settled captures use near 0.05 and far 2048 with inverted depth. The early infinite-far lead was not reproduced as a persistent SR condition; do not add infinite-depth flags blindly.
Frame time	Frame intervals are supplied in milliseconds. Capture-time stalls can trigger intentional resets; exclude those windows from stability comparisons.
Masks	Nearest history lookup and thresholded composition merit a targeted A/B after the jitter fix. Static mask probes reject only a small part of the frame, so blanket whole-frame rejection was not observed.
Frame generation	Its orientation conversion and presentation resource lifecycle are separate. The new audit does not certify FG output, pacing or latency.

Keep any UI description precise: a higher FSR quality preset provides more input samples, while sharpness only changes edge contrast. Neither is a replacement for correctly aligned temporal data.

DLSS: the floating-line reproduction



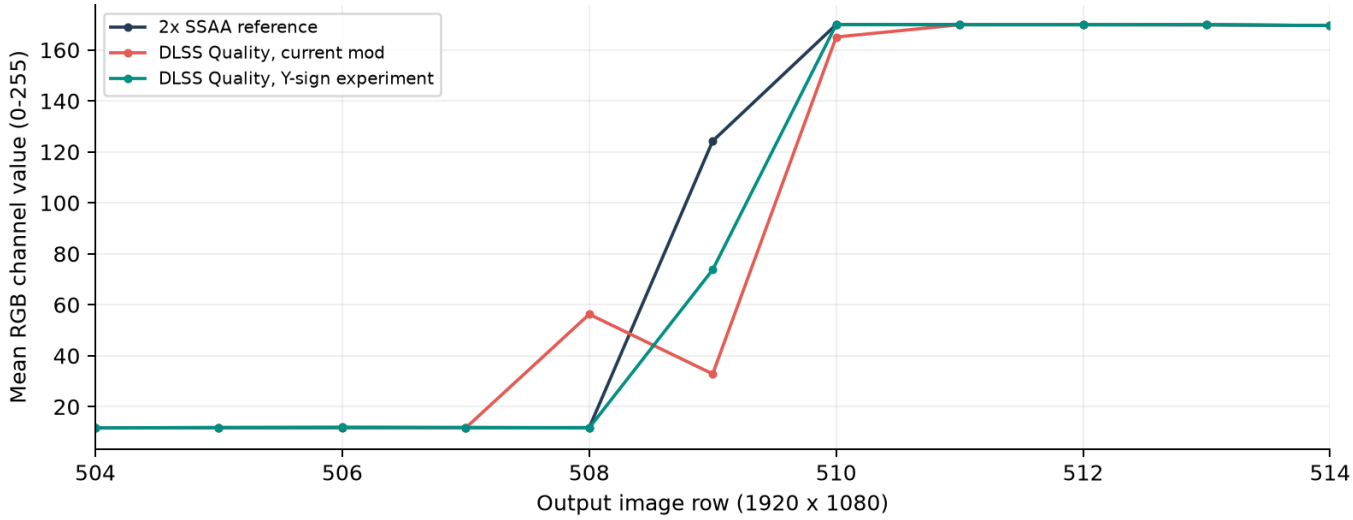
Each source crop is 90 x 32 pixels around an isolated block top. Nearest-neighbor enlargement. Top: 2x SSAA. Middle: current DLSS Quality at sharpening 0. Bottom: only the submitted Y jitter sign is inverted. The thin separate row above the middle block edge disappears in the diagnostic output.

Measured evidence: averaging 96 frames over columns 940-979, row 508 is 56.27/255 and row 509 is 32.85/255 in the current output: a 23.42-level downward step before entering the block. The sign-corrected test rises through those rows instead (11.67 to 73.75).

The bright row/dark gap is a reproduced form of the user's floating-line report. It is a reconstruction artifact in this fixture, not a new block or physical line. Different scenes may contain other kinds of floating lines; this experiment does not establish a cause for every possible instance.

Primary solution: correct the SR jitter convention. The artifact occurs with added sharpening at zero, so blaming the mod's optional sharpening alone is not supported.

DLSS: profile and sample coverage



Mean display-channel profile across the same block edge. The non-monotonic step in the current DLSS output is distinct from the smooth rise in the reference and sign-only experiment.

D1 - phase coverage: DLSS takes the shared TAA phase setting, normally 16, instead of deriving coverage from its queried render dimensions. NVIDIA recommends a base of 8 multiplied by the output/input area ratio. [N1]

DLSS mode	Current cycle	Recommended starting count
Quality	16	18
Balanced	16	24
Performance	16	32
Ultra Performance	16	72

The mod's TAA control sanitizes its values to 8/16/32, so exposing that same control cannot correctly represent a 72-phase Ultra Performance sequence. Use a separate DLSS automatic calculation; do not change TAA behavior merely to compensate for DLSS.

A practical implementation is ceil(8 * outputWidth * outputHeight / (renderWidth * renderHeight)), using the actual optimal settings. Match the existing reset behavior when dimensions or sequence length change.

DLSS: phase experiments and quality limits

Matched 144-frame test	Floor SD	Thin SD	Interpretation
Quality / 16 phases	0.080	0.292	Y convention unchanged
Quality / 18 phases	0.095	0.324	Y convention unchanged
Ultra Performance / 16 phases	0.255	0.416	Y convention unchanged
Ultra Performance / 72 phases	0.332	0.772	Y convention unchanged

Changing the number of samples is not the same as correcting their orientation. These matched tests isolate phase count while retaining the current Y convention. A longer cycle is required by the integration guidance but is not proven here to be a standalone flicker cure. Do not promise that “72 phases fixes DLSS.”

1080p preset	Floor SD	Thin SD	Glass SD
Quality	0.079	0.271	0.254
Balanced	0.081	0.275	0.251
Performance	0.290	0.871	0.527
Ultra Performance	0.230	0.444	0.294

The Performance capture varies more than Ultra Performance in some regions. That does not make Ultra Performance better: blur, aliasing distribution and missing detail can lower a variance score. At 1080p it reconstructs from only 640 x 360, one ninth of the output pixel count.

The pinned Streamline contract also requires matching frame tokens/resource extents and unjittered matrices. Current static vectors are finite and near zero, and successful native calls were observed. Those checks establish a functioning path; they do not certify its visual quality. [N2] [N4]

DLSS: other tested cases and fixes



DLSS Quality natural-terrain motion crops. The sequence probes fine foliage and newly exposed surfaces. It is not a synchronized reference comparison.

Selected-block views and first-person rendering were captured in Creative mode with a held pickaxe and visible HUD. The detached-line fixture is a separate stationary spectator test, so its reproduction does not depend on the selection outline or the hand.

The hand retains visibly jagged edges in the first-person capture. The native OFF control is also jagged there. This is a remaining presentation-quality gap, not evidence that DLSS invented the aliasing. If desired, add a dedicated hand-AA pass or include the hand in a carefully defined temporal path with matching depth/motion; verify weapon swing, occlusion and UI isolation.

After the primary fix: test the current-color bias mask on transparent/moving content, verify camera-matrix convention during vertical and diagonal movement, and compare a clean driver profile with any user-forced DLSS overrides. A new model or runtime should be a separate controlled experiment, not an assumed cure for the demonstrated sign error.

NVIDIA's Vulkan sample makes its clip-space and matrix conversions explicit. Use it as a coordinate-contract cross-check, while respecting Minecraft's actual surface orientation. Copying a sign from another renderer without tracing the image convention can introduce a second error. [N3]

CROSS-METHOD CONFIRMATION

Resolution and settings sensitivity

Quality floor SD	720p current / test	1080p current / test	1440p current / test
FSR2	0.386 / 0.249	0.321 / 0.189	0.263 / 0.142
FSR3	0.269 / 0.178	0.222 / 0.148	0.160 / 0.111
DLSS	0.110 / 0.070	0.079 / 0.053	0.059 / 0.043

The table uses matched sign tests, not the earlier preset survey. Higher output resolution does not make a coordinate mismatch disappear. The experiment is an isolated alignment check; it is not a final optimization benchmark.

Setting tested	What the evidence supports
Added sharpening 0 vs 25%	The line artifact already exists at 0. Added edge contrast can make unstable fine detail more visible; apply sharpening only after reconstruction is correct.
Mipmap bias OFF vs automatic	More aggressive texture sampling can expose high-frequency instability. It does not repair temporal coordinates. Retain a conservative option and tune per material if necessary.
Masks on vs off	Disabling masks did not eliminate the opaque-floor symptom. Global disabling is not a valid general fix for transparent/moving content.
Quality vs Ultra Performance	Quality gives substantially more input detail. Ultra Performance may reduce a variance metric through lost detail; keep the missing-information tradeoff explicit.

AMD recommends considering content-specific mip bias for textures prone to temporal aliasing. A global negative bias should be judged against shimmer and texture readability together. [A2]

RENDERING STRESS COVERAGE

Lifecycle tests and non-findings

Test	Observed outcome	Limits
Horizontal / vertical sweeps	Captured for OFF, FSR2, FSR3 and DLSS, including slow/fast paths.	Tick-driven camera paths stress reconstruction but are not a motion-compensated ground-truth comparison.
Entities / chest activity	96-frame captures for each upscaler with server time running.	Does not certify every entity type, animation or modded material.
Timed teleport during capture	One reset is observed in each method: FSR2 frame 39, FSR3 frame 19, DLSS frame 39 of their respective 120-frame sequences.	Different polling phases change when the command reaches the game. Reset alignment is assessed against each captured scene transition.
Resize 1080p to 720p and back	All paths recovered. New-size DLSS dispatch begins with reset=true.	Minecraft emitted recoverable swapchain-out-of-date warnings. These were not crashes.
Static GPU motion probes	Finite values with very small stationary magnitudes; representative FSR2 maximum about 0.000022 input pixels.	Near-zero stationary vectors cannot validate moving-object reprojection or disocclusion depth.
Masks / native success	Static rejection fractions were small; native upscaling status remained successful during the retained quality sequences.	Success codes do not imply correct images. No universal artifact-free claim follows.

No persistent black output, vertical inversion of the final SR image, whole-frame invalid motion field, perpetual static reset, or reproducible resize crash was established. The early capture-helper serialization failure was corrected in test code and excluded; it is not a production-mod finding.

RECOMMENDED NEXT WORK

Implementation plan and acceptance checks

Priority	Concrete change	Verify before acceptance
P1 / shared	Correct native SR Y jitter at the FSR and DLSS boundaries; name the raster/SDK conventions.	Floating-line profile loses its bright-row/dark-gap pattern; matched variance improves without broadening edges or adding motion trails.
P1 / DLSS	Derive automatic phase counts from actual render/output dimensions. Decouple from the capped TAA control.	18/24/32/72 nominal coverage; reset once on sequence changes; combined sign+phase tests in all explicit presets.
P2 / transparency	Capture mask buffers per frame; test fractional, depth-aware history lookup and material-based reactivity.	Less glass/particle shimmer and no long trails. Compare mask footprint to actual changed/translucent pixels.
P2 / validation	Verify motion direction and pixel units with controlled horizontal, vertical and diagonal object/camera translations.	Reprojection lands on the same surface; depth selects correct foreground at disocclusions; no stale velocity on newly spawned objects.
P3 / presentation	Evaluate a dedicated hand-AA solution and conservative texture/sharpen defaults.	Clean hand swings and world/hand boundaries; HUD/text stay stable and sharp.

Use the corrected diagnostic cases as regression references, not a demand for zero numerical variance. Require visible improvements on solid silhouettes and fine geometry, preserve detail against SSAA, and compare moving clips for ghost trails. A blanket 0.000 variance target would incorrectly reward frozen or blurred output.

Package a test JAR and repeat in a normal Fabric installation as well as Gradle runClient. Recheck missing/unsupported DLSS fallback, world entry/exit, GUI scale, fullscreen changes, resource reload and memory pressure. Those packaging/compatibility checks are follow-up acceptance work, not completed by this report.

APPENDIX / RETAINED LOCAL EVIDENCE

Evidence map and reproducibility

The retained inventory contains 151 capture runs and 8,879 output PNGs, plus the diagnostic input-image series. Counts include setup and exploratory runs; the key conclusions use the specific matched windows identified in the report.

Evidence	Location relative to repository
Full output images and run metadata	run/upscaler-audit-20261024/screenshots/ and audit-*.json
Per-dispatch input records	run/upscaler-audit-20261024/inputs-*.jsonl
GPU probes	run/upscaler-audit-20261024/qa-motion-*.json and qa-mask-*.json
Metrics / edge profiles	build/upscaler-audit-20261024/metrics.json, profiles.json and input-jitter-correlation.json
Capture requests / inventory	build/upscaler-audit-20261024/*-plan.json, requests.jsonl and capture-inventory.json
Audit code / analysis	build/upscaler-audit-20261024/src/, client.gradle, final_analysis.py and report.py
Build and provenance	build/upscaler-audit-20261024/launch*.log, environment.json and source-sha256.json

The initial scene called “ledge” was occluded by the existing fixture and is not used to claim isolated-silhouette behavior. The corrected “silhouette” scene is the source of the floating-line and edge-profile evidence. Initial teleport captures could miss the transition; the 120-frame “reset-exact” tests arm capture before sending the command.

Scope limits: one GPU/driver and a development client; no AMD/Intel hardware, OpenGL/Sodium/Iris compatibility matrix, DLAA, DLSS developer accumulation mode, DLSS model-override certification, underwater/special-dimension certification, or new FSR3 frame-generation certification. Passing these scene tests cannot prove every Minecraft material or future mod interaction artifact-free.

APPENDIX / PRIMARY RESEARCH

Sources and code reviewed

[A1] AMD FSR2 2.3.3 integration manual

https://gpuopen.com/manuals/fidelityfx_sdk/techniques/super-resolution-temporal/

[A2] AMD FSR3 3.1.4 upscaler integration manual

https://gpuopen.com/manuals/fidelityfx_sdk/techniques/super-resolution-upscaler/

[A3] AMD FidelityFX API / FSR3.1 integration presentation

https://gpuopen.com/presentations/2024/FidelityFX_Super_Resolution_3-1_Release-Overview_and_Integration.pdf

[N1] NVIDIA DLSS Programming Guide, 31 March 2026

https://raw.githubusercontent.com/NVIDIA/DLSS/main/doc/DLSS_Programming_Guide_Release.pdf

[N2] NVIDIA Streamline DLSS guide, pinned v2.14.1

https://github.com/NVIDIA-RTX/Streamline/blob/v2.14.1/docs/ProgrammingGuideDLSS.md

[N3] NVIDIA Vulkan Streamline reference sample

https://github.com/nvpro-samples/vk_streamline/blob/main/main.cpp

[N4] NVIDIA Streamline constants, pinned v2.14.1

https://github.com/NVIDIA-RTX/Streamline/blob/v2.14.1/include/sl_consts.h

Primary sources accessed 24 October 2026. The pinned Streamline guide matches the tested integration generation; live AMD manuals and the NVIDIA guide provide the expected input/quality contract. Vendor images from different games are not used as a misleading pixel reference.

Principal implementation locations

ModernMinecraftHooks.prepareTemporalJitter; VulkanSceneTemporalController.prepareFrameJitter/configureProjection; VulkanSceneFsrController; VulkanSceneDlssController; VulkanFsrTemporalMaskRenderer; VulkanTemporalColorRejectionRenderer; shaders/include/temporal_motion.glsl; shaders/post/fsr_temporal_masks.fsh; native/fsr_bridge/src/salts_fsr_bridge.cpp; native/dlss_bridge/src/salts_dlss_bridge.cpp.

The research establishes the integration requirements. The screenshots, numeric comparisons and diagnostic sign experiments are original findings from this mod. Recommendations concerning remaining motion/material behavior are explicitly distinguished from completed fixes.